

WHY ENGLISH DOESN'T EXTRACT LEFT BRANCHES (YET)

Brett Reynolds 

Humber Polytechnic & University of Toronto

brett.reynolds@humber.ca

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ABSTRACT

English resists separating pre-head dependents from the head of an NP: **Whose did you read book?* is sharply unacceptable. Rather than positing a universal constraint, this paper asks what English does instead – and measures it. The explanation has three strands: *distributional* (fused-head construals and workarounds like pied-piping and *big mess* are frequent; configurations resembling left-branch extraction are absent), *psycholinguistic* (fused-head parsing commits listeners early, making reanalysis costly), and *functional* (the workaround family already does the semantic work that extraction would do). Corpus rates from Universal Dependencies confirm: workarounds are entrenched, determiner–head packaging is tight, and discontinuity is near-zero. A Bayesian learner has no positive evidence for separation; the posterior stays near zero and the gap perpetuates itself. Absence of evidence, when the search is thorough, is evidence of abstinence. And abstinence, unlike impossibility, can end.

Keywords: Left-Branch Condition, Bayesian acquisition, distributional learning, functionalism, English syntax, corpus linguistics.

I INTRODUCTION

English makes no semantic mystery of *whose* in (1), but the stranded-noun variant (1b) is sharply unacceptable:

- | | | | |
|-----|----|----------------------------------|---------------|
| (1) | a. | <i>You read whose book?</i> | [in-situ] |
| | b. | <i>*Whose did you read book?</i> | [LBE] |
| | c. | <i>Whose book did you read?</i> | [pied-piping] |
| | d. | <i>Whose did you read?</i> | [fused-head] |

As an echo question, the in-situ form (1a) is fine; for ordinary information-seeking questions English pied-pipes the noun (1c). What fails is fronting *whose* while leaving *book* behind (1b). When the noun is omitted, *whose* serves as a fused-head NP (1d), with no overt noun expected.

The point is general: English resists splitting a pre-head dependent from the noun it depends on. Compare:

- | | | | |
|-----|----|--------------------------------------|---------------|
| (2) | a. | <i>She earned how much money?</i> | [in-situ] |
| | b. | <i>*How much did she earn money?</i> | [LBE] |
| | c. | <i>How much money did she earn?</i> | [pied-piping] |
| | d. | <i>How much did she earn?</i> | [fused-head] |
-
- | | | | |
|-----|----|--|-----------|
| (3) | a. | <i>I don't think you liked his food.</i> | [in-situ] |
|-----|----|--|-----------|

- b. * *His, I don't think you liked food.* [LBE]
- c. *His food, I don't think you liked.* [topicalised NP]
- d. *His, I don't think you liked.* [fused-head]

These are instances of what is traditionally called **LEFT-BRANCH EXTRACTION (LBE)**: the fronting of a pre-head dependent – possessor, quantifier, degree phrase – while the head remains in situ. Calling it **LEFT-BRANCH EXTRACTION** names the missing behaviour without explaining why English keeps its left branches on a short leash.

Why is English so resistant to LBE? Is the gap a universal structural prohibition – a constraint that holds across languages – or an English-specific pattern emerging from how English packages its nominals?

The standard move in generative syntax is to posit a constraint: the Left-Branch Condition, or some descendant of it (Ross, 1967). But stipulating a ban doesn't explain why the ban exists, or why English obeys it so strictly when other languages appear more permissive. A constraint is a good diagnostic but a poor causal story. For other “island types” (a useful descriptive grouping, if not a natural one), functionalist accounts have made progress: Culicover et al. (2022) tie unacceptability to processing complexity (cf. Gibson, 1998; Hawkins, 2004, on dependency locality); Cuneo and Goldberg (2023) show that island effects correlate with discourse function. But LBE remains unexplained. The Left-Branch Condition doesn't satiate like other islands, and no discourse-functional account of the determiner–head bond has been proposed.

Instead of asking “what rule blocks LBE?”, this paper asks: what does English do instead? The explanation that emerges has three strands – distributional, psycholinguistic, and functional – that jointly explain why the gap is stable.

The **DISTRIBUTIONAL** strand is straightforward. English makes heavy use of **FUSED-HEAD** nominals – constructions where a determinative or modifier realises the head function, leaving no overt noun. And English has a family of productive constructions that express degree modification and fronted-determiner dependency formation without separating determiner from head: pied-piping (*whose book*), the *big mess* construction (*how good a teacher*), *of*-insertion (*how big of a risk*), and predeterminer modifiers (*what a mess*). These workarounds are frequent and entrenched (cf. Bybee, 2010, on entrenchment); hypothetical LBE is not. The corpus data shows that workarounds occur at measurable rates while LBE-like configurations are essentially absent.

The **PSYCHOLINGUISTIC** strand follows from the first. When a speaker fronts *whose*, a listener's first parse is likely to treat it as a complete NP (*whose* = ‘whose one’), not as a displaced dependent. Fused-head productivity does double duty: it provides an alternative to LBE in production, and it biases comprehension toward a fused-head construal. An LBE parse would require costly reanalysis.

The **FUNCTIONAL** strand closes the loop. The workaround family does the semantic work that LBE would do. There is no communicative pressure favouring a new construction; the equilibrium perpetuates itself.

The strategy is to measure, not judge. The stars in (1)–(3) have already done plenty of judging; the question here is what English does with its time instead. Rather than collecting acceptability ratings for starred sentences, this paper estimates rates: how often do these workaround constructions occur, how tight is determiner–head packaging, and how rare are configurations that would require an LBE parse? The aim is English-internal. The data comes from English usage, and the explanation invokes English-specific distributional facts. This framing brackets cross-linguistic comparison.

Slavic languages are often said to “allow LBE”, but whether the Slavic pattern and the English gap instantiate the same structural configuration is far from clear. LEFT BRANCH is a descriptive class tied to a particular phrase-structure analysis – not a valid comparative concept (Haspelmath, 2010). Following Croft (2001), this paper treats the English pattern as English-specific, without assuming that discontinuous nominals in case-rich, article-less languages are commensurate.

The logic is Bayesian: if learners have no positive evidence for determiner–head separation, they won’t acquire it. The gap perpetuates itself not because grammar forbids LBE, but because the discourse community’s use of fronting is restrained. The familiar aphorism says that absence of evidence is not evidence of absence. But when the opportunity set is large and the search is thorough, absence of evidence *is* evidence – evidence of abstinence. Call it the dog-that-didn’t-bark inference, but with a denominator. And abstinence, unlike impossibility, can end.

2 THE CONSTRUCTIONS

Before operationalising anything, it helps to see the constructions tracked here. Table 1 shows the main patterns, with examples. In the table, “int./rel./excl.” means interrogative, relative, or exclamative.

TYPE	PATTERN	EXAMPLES
Pied-piping	interrog./rel./excl. + N	<i>whose book, which car</i>
Fused-head	interrog./rel./excl. alone (headless)	<i>whose is better, which did you choose</i>
<i>big mess</i>	degree + Adj + <i>a</i> + N	<i>how good a teacher, so big a deal</i>
<i>of</i> -insertion	degree + Adj + <i>of a</i> + N	<i>how big of a risk, too good of a deal</i>
Predeterminer modifier	modifier + <i>a</i> + N	<i>what a mess, such a deal</i>

Table 1: English constructions relevant to LBE.

The label *big mess* is informal but apt: these are precisely the places English tidies up the semantics while refusing to untie determiner from head.

What unites these patterns is that they do the semantic work that hypothetical LBE would do – expressing degree, quantity, or possession in interrogatives and exclamatives – while keeping determiner and head contiguous. These are not marginal constructions: Sommerer (2022) shows that *big mess* patterns like *so great a desire* were far more frequent in Early Modern English than they are today, and Van Eynde (2007) provides a formal HPSG account of their synchronic structure.

Just how tight is determiner–head packaging? Jackendoff (2025) argues that full parenthetical clauses can’t interrupt, e.g.:

- (4) * *The, I think, report was leaked.*

A search of our corpora (see Section 4) for determiner–head arcs with intervening punctuation, followed by manual inspection for parenthetical supplements, yields no clear cases of phrase-sized interruptions.¹ Jackendoff’s explanation is prosodic: parentheticals require Intonational Phrase boundar-

¹The search filtered det-arc candidates with intervening punctuation and manually inspected for appositives, non-restrictive relatives, and parenthetical clauses. Apparent hits were lists or punctuation inside modifier phrases, not genuine det–supplement–head sequences. Hesitation markers like *the, uh, report* may occur in speech, but these are disfluencies, not grammaticalised supplements; the corpora here are predominantly written.

ies, and a determiner and its head form a tight prosodic unit – typically a clitic group or single Phonological Phrase – with no internal boundary where a supplement could attach (Jackendoff, 2025, pp. 33, 42); for the prosodic hierarchy that makes this prediction, see Selkirk (1984) and Nespor and Vogel (2007). If the bond resists even parenthetical phrases, extraction is out of the question.

This parsing bias is noted by Culicover et al. (2022, p. 24): a clause-initial *which*, *what*, or *whose* form is processed by default as a phrasal argument with an elided nominal head – that is, as a fused-head NP. The result is a garden-path effect when the listener encounters a bare noun with no determiner. English fused-head productivity, then, does double duty: it provides an alternative to LBE in production, and it biases us against an LBE construal in comprehension.

3 A BAYESIAN PERSPECTIVE ON ACQUISITION

The acquisition question here isn't whether a learner can entertain some abstract possibility like "a pre-head dependent separates from its head". It is what analysis the learner assigns to the kinds of strings English actually provides (cf. Tomasello, 2003, on usage-based acquisition). In the domain at issue, English input supplies two analyses that are both frequent and that both keep determiner and head contiguous: headed NPs with pied-piping (*whose book*, *how much money*) and fused-head NPs where the fronted form itself realises head function (*whose did you read?*, *how much did she earn?*). A hypothetical LBE analysis would add a third option: the fronted item is a displaced dependent whose head noun will appear later without it.

What makes absence informative here is that the opportunity set is large. Fronting is everywhere in English input: open interrogatives, relative clauses, exclamatives, topicalisation, and focus fronting. A learner repeatedly encounters contexts where an LBE-style analysis would be usable and diagnosable – contexts where a fronted form could, in principle, be parsed as a displaced dependent of a later noun. So the learner doesn't face an undifferentiated "infinite hypothesis space" in which negatives are unlearnable. They face a neighbourhood of analyses in which one extension is a natural next step but is systematically absent from exactly the contexts where it would be licensed.

Given this setup, a Bayesian framing isn't window dressing but a compact way of stating the inference problem (cf. Goldwater et al., 2009; Xu & Tenenbaum, 2007, for Bayesian approaches to word learning and segmentation). The intuition is simple: beliefs should reflect evidence. A learner who has encountered a thousand opportunities for LBE and zero instances has strong evidence that LBE doesn't happen. Not proof – you can't prove a negative – but strong evidence. The more opportunities, the stronger the evidence. This is what Bayesian reasoning formalises: your confidence in a hypothesis should track how well it predicts what you actually observe (Perfors et al., 2011; Yang & Piantadosi, 2022). If your grammar predicts that LBE should sometimes occur, but it never does, that grammar fits the data poorly. A grammar that predicts determiner–head contiguity fits better.

Two distributional facts push hard in the same direction. First, fused-head nominals are genuinely productive in English. That means that when a learner hears a fronted *which/what/whose* in an interrogative or relative, the parse 'this is already a complete NP' has substantial prior support from experience. Second, the headed alternative that English also supplies is pied-piping, which again preserves determiner–head contiguity. In that landscape, an LBE analysis isn't merely unattested; it is a structurally costlier commitment. It asks the learner to treat the same surface cue (fronting) as licensing a dependency whose downstream signature is determiner–head discontinuity with an overt nominal appearing later – exactly the configuration that competes with the highly entrenched fused-head

parse. The comprehension consequences noted in Section 2 follow directly: fused-head productivity does double duty, offering a production alternative and biasing comprehension away from an LBE construal.

LBE is *on the table* for an English learner in this sense: it is assigned non-zero prior probability because it is a locally conceivable extension of a richly attested strategy, not an alien operation with no foothold in the grammar (cf. Perfors et al., 2011, on hypothesis-space constraints). As an analogy for what it looks like when the input supplies no nearby scaffolding, consider productive noun incorporation of the Mohawk type, which involves building a lexical noun root into the verb word itself, yielding a single finite predicate with a non-referential incorporated nominal:

- (5) *Kanekwarínyu wa'-k-akya'tawi'tsher-ú:ni*
 it.dotted FACT-1SG-dress-make
 'I dress-made a polka-dotted one.' (Mohawk; Mithun 1984)

English can of course express the propositional content, but it doesn't supply a productive schema in which incorporation systematically alternates with a transitive clause in a way that tracks the specific/non-specific contrast typical of incorporation systems. Lexicalised V-N compounds like *babysit* or *spoon-feed* don't provide the same constructional logic. In Bayesian terms, a learner may be able to represent such a grammar in principle – incorporation isn't logically impossible – but the prior mass assigned to it is vanishingly small because English input provides no scaffolding constructions from which it could be inferred. LBE is different: the analogues (fronting, pied-piping, fused-head construals) are abundant. What the learner acquires, then, is not a rule but a norm: we don't do that.

That last clause matters. What is acquired here isn't an abstract theorem about "English" but a community norm about packaging. If a learner has spent years hearing determiners and prenominal dependents appear tightly integrated with their heads, and has heard fronting realised overwhelmingly through pied-piping or fused-head NPs, then a string like **Whose did you read book?* violates an entrenched community expectation. The felt unacceptability is the experiential signature of that expectation being breached: "that's not how we talk." This kind of account doesn't require a derivational ban. It requires only that learners infer domain restrictions on otherwise productive schemas from distributional evidence.

A community-norm framing also makes room for change without weakening the synchronic explanation. Learners don't treat all input as coming from one homogeneous source – they notice that different people talk differently. Teenagers talk differently from grandparents; Twitter is different from academic prose. A learner tracking "how often does this structure occur?" is really tracking "how often does this structure occur *among people like X?*" For mainstream English speakers, the answer for LBE is: never, as far as I can tell. But if a subgroup started using LBE – if it became a marker of some community – a learner could notice that and track it separately. The account is compatible with language change: what shifts is not a universal constraint but which community's patterns the learner is tracking. This is also why pooling across corpora makes sense in the analysis below: the learner is effectively averaging over sources, weighting each by how much data it provides.

This perspective also explains a puzzling experimental finding. Syntactic SATIATION is the tendency for repeated exposure to make initially unacceptable sentences more acceptable (Snyder, 2000). Many island violations show satiation: hear them enough times in a lab, and they start to sound better (Lu et al., 2024). But LBC violations don't satiate. Lu et al. (2024), in a meta-analysis of 25 experiments, found no satiation effect for left-branch extraction. Why not? On the present view, the

answer is simple: a lifetime of evidence beats a few minutes in a lab. If you’ve heard tens of thousands of determiner–head sequences and zero LBE cases, hearing a handful of **Whose did you read book?* in an experiment won’t budge your expectations much. The evidence against LBE is too overwhelming.

This framing makes the empirical question explicit. If the Bayesian account is right, we should find:

1. Workaround constructions at non-trivial rates relative to their opportunities.
2. LBE-like configurations at near-zero rates relative to their opportunities.
3. Tight determiner–head packaging, with little or no intervening material.

The next section describes how to measure these rates.

4 DATA AND METHOD

The predictions above are empirical, and testing them requires measuring rates – not raw counts, but relative frequencies. Saying “the *big mess* construction is frequent” means nothing without a denominator: frequent relative to what?

Corpus data can test this directly. Unlike acceptability judgments, which ask whether a structure *could* occur, corpus rates tell us what *does* occur – and so what learners are exposed to. The goal is to estimate, with uncertainty, the rate of each construction given its opportunity set.

The opportunity set differs by construction. For the *big mess* construction (*that big a mess, how good a teacher*), the denominator is: how many configurations involve degree modification of an adjective with an indefinite NP? The rate is *big mess* tokens divided by that opportunity count. For fused-head construals (*whose* as a complete NP), the denominator is: how many fronted *which/what/whose* forms occur with an overt head versus without one? For LBE-like configurations: how many determiner–head arcs are there, and how many are discontinuous?

LBE itself can’t be searched in corpora – we can’t find tokens of **Whose did you read book?* because speakers don’t produce them. Instead, the analysis operationalises what LBE *would* look like if it existed: a determiner separated from its head by intervening material (a verb, a crossing dependency). If such configurations are near-zero, the corpus provides no evidence from which a learner could acquire determiner–head discontinuity.

WHAT AN LBE TOKEN WOULD LOOK LIKE

If English permitted LBE, a sentence like **Whose did you read book?* would require a dependency annotation where *whose* bears a DET relation to *book*, but the arc crosses the finite verb *read*. Schematically:

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Whose did you read book ?
|→→→→det→→|
1      2    3    4    5    6

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The DET arc from position 1 to position 5 spans four tokens and crosses a verb. This is the signature: a determiner–head arc in the fronting domain (interrogative, relative, or exclamative fronting) that either (i) spans an intervening verb, or (ii) crosses another dependency arc. If English speakers productively used LBE, such configurations would appear at non-trivial rates in the fronting subset. They don’t. The search below operationalises exactly this: fronted determiners whose head noun is separated by a verb or by a crossing arc.

The data comes from Universal Dependencies (UD; Nivre et al., 2020), a cross-linguistic annotation standard that provides consistent syntactic structure across corpora. The UD slice includes thirteen English corpora spanning edited and unedited registers: newspaper, web text, blog posts, dialogue, learner English, and more. UD is used diagnostically, not as a theory of English structure; every pattern is checked against standard terminology. The search procedure is high-precision by design: it is tuned to produce clean denominators rather than exhaustive recall.

Workaround operationalisations are defined in dependency and order terms. The *big mess* construction is operationalised as degree modifier + adjective + *a/an* + noun in that order, as in *how good a student*. The *of*-insertion variant is parallel but includes *of*, as in *how big of a risk*. The predeterminer modifier cluster captures *such a*, *what a*, and *many a*.

To approximate fused-head availability, the analysis uses UD DET vs PRON tags for *which*, *what*, and *whose*. When UD tags one of these words as PRON, it typically reflects a headless (fused-head) construal; when it tags the same word as DET, the word is in determiner function with an overt head. This is a proxy for headless nominal construals, not a claim about lexical category.

Determiner–head packaging is measured by distance (in tokens), with a threshold of $k = 4$: arcs longer than this are flagged as potential discontinuities. The analysis also tracks whether a verb or crossing arc intervenes, as in *the students who arrived yesterday* (verb-intervening) or *the book, which I recommended, on syntax* (crossing). Arcs with verb or crossing interveners are flagged as LBE-like candidates. The threshold was validated (§7) and treated as a sensitivity parameter.

5 WHAT WE'RE MEASURING AND HOW

The goal is to estimate rates – not to pronounce sentences grammatical or ungrammatical, but to ask: how often does each construction occur, given its opportunities? Four rates matter:

1. How often do workarounds (*big mess*, *of*-insertion) occur when the opportunity arises?
2. How often do speakers use fused-head construals (*whose* alone) versus headed ones (*whose book*)?
3. How tight is determiner–head packaging? How far apart can they get?
4. How often do we see the kind of discontinuity that LBE would produce?

Estimating a rate from a single corpus is straightforward: count the hits, divide by opportunities. But corpora differ in size, and small corpora give noisy estimates. If one corpus has 2 hits out of 10 opportunities (20%) and another has 200 out of 1,000 (also 20%), we should trust the second estimate more.

PARTIAL POOLING handles this (Gelman & Hill, 2007; McElreath, 2020). The idea: let the corpora inform each other. If most corpora cluster around 5% but one tiny corpus shows 50%, that extreme probably reflects noise, not a real difference. Partial pooling pulls extreme small-sample estimates toward the group average while letting large corpora speak for themselves. The result is CREDIBLE INTERVALS – ranges that tell you where the true rate probably lies. A 95% credible interval means: given the data, there's a 95% probability the true rate is in this range. (This is what many people *want* confidence intervals to mean; credible intervals, unusually, do mean what they sound like they mean.)

For rare phenomena – like LBE-like configurations – the question shifts from “what's the rate?” to “how high could the rate possibly be?” If you search 1,000 opportunities and find nothing, the true rate is almost certainly below 0.3%. The rule of thumb: with zero hits, the 95% upper bound is roughly 3 divided by the number of opportunities. This is the sense in which absence of evidence

becomes evidence of absence: not proof of impossibility, but a ceiling on how common the phenomenon could be.

For readers who want to see what the model actually looks like, here it is:

$$\begin{aligned} k_j &\sim \text{Binomial}(N_j, \theta_j) \\ \theta_j &\sim \text{Beta}(\bar{\alpha}, \bar{\beta}) \end{aligned}$$

That’s it. Each corpus j produces k_j hits out of N_j opportunities, with its own rate θ_j . The rates are drawn from a shared Beta distribution with parameters $\bar{\alpha}$ and $\bar{\beta}$ – this is the partial pooling. The Beta distribution is just a way of representing uncertainty about a rate; its two parameters control where the distribution is centred and how spread out it is. When $\bar{\alpha}$ and $\bar{\beta}$ are estimated from the data, extreme corpus-level rates get pulled toward the group mean. With a flat prior (Beta(1,1), which says “any rate is equally likely before seeing data”), the posterior on each θ_j reflects mainly the corpus’s own evidence, modulated by what the other corpora suggest is typical. The results are robust: using different reasonable priors gives essentially the same answers, because the sample sizes are large enough that the data dominate (Gelman et al., 2013, ch. 5).

Figure 1 shows what partial pooling does in practice. The open circles are raw corpus rates (bigger circles = more data); the filled circles are the pooled estimates with uncertainty intervals. Notice how small corpora get pulled toward the group, while large corpora barely move. Figure 2 shows why: bigger samples give tighter estimates. This is the payoff of partial pooling – it lets small corpora contribute without letting their noise dominate.

6 RESULTS

I went looking for LBE. Seven corpora, 86,098 determiner–head arcs, 1,228 opportunities in the fronting domain where LBE would be diagnosable. Nothing. If English has an LBE, it has learned to avoid annotated corpora with professional discipline. The workarounds are there; the gap is real.

Across the UD corpora, there are 15 *big mess* tokens and 4 *of*-insertion tokens against 85 degree+adjective+*a/an*+noun opportunities. The predeterminer modifier cluster yields 144 tokens: *such a, what a, many a*. The denominator is 21,580 *a/an* determiner–head arcs. The fronted-form proxy yields 1,228 fronted determiner arcs and 6,429 PRON-tagged tokens overall, so the fused-head option is well attested in usage even in a conservative UD proxy.

Figure 3 and Table 2 summarise the workaround family by construction. The figure shows rates by corpus as points with a distributional summary, which makes the between-construction contrasts visible without hiding register variation. Figure 4 unpacks this by corpus, showing how register variation interacts with construction choice.

Determiner–head packaging is tight. Table 3 summarises the raw counts for LBE-like configurations. The critical row is the last: out of 1,228 fronted determiner arcs, only 2 involve a verb or crossing – the configuration that would signal LBE. Under the partial-pooling model, the 95% upper bounds on the fronted determiner verb/cross rate are below 0.5% in every corpus, typically around 0.3–0.4%.

The two fronted determiner hits are *What a moving van!* (CHILDES) and *what a mindblowing servicing* (EWT). Both are exclamatives where a participial adjective (*moving, mindblowing*) triggers

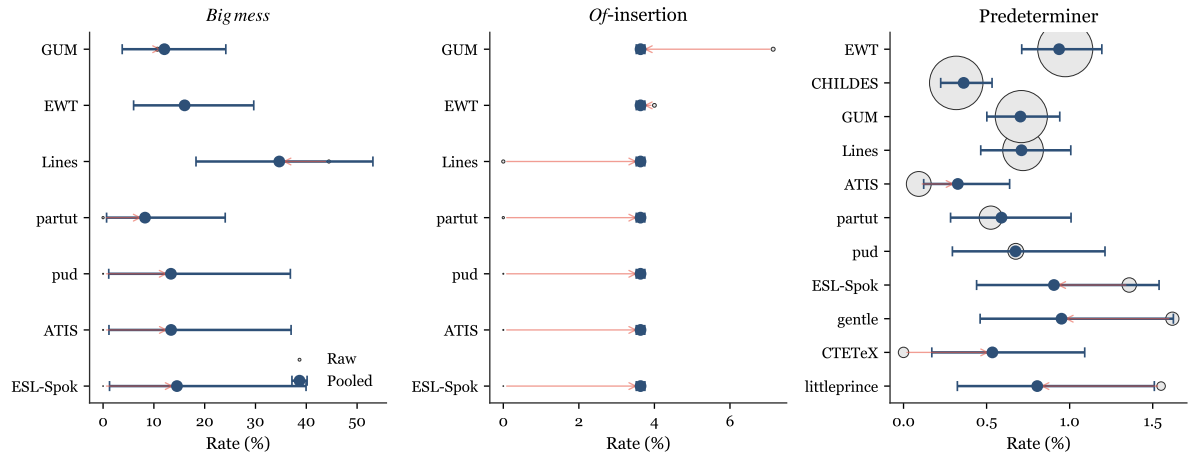


Figure 1: Partial pooling in action. Open circles show raw corpus rates (sized by sample); filled circles show pooled posterior means with 95% intervals. Arrows indicate shrinkage direction.

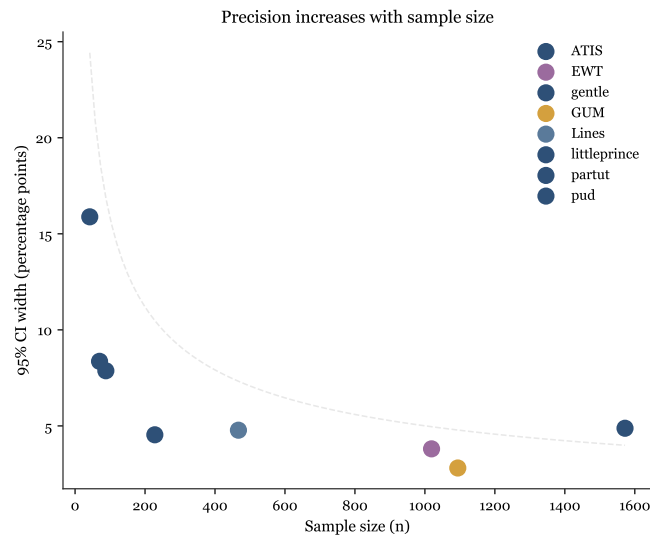


Figure 2: Credible interval width decreases with sample size, following the expected $1/\sqrt{n}$ relationship.

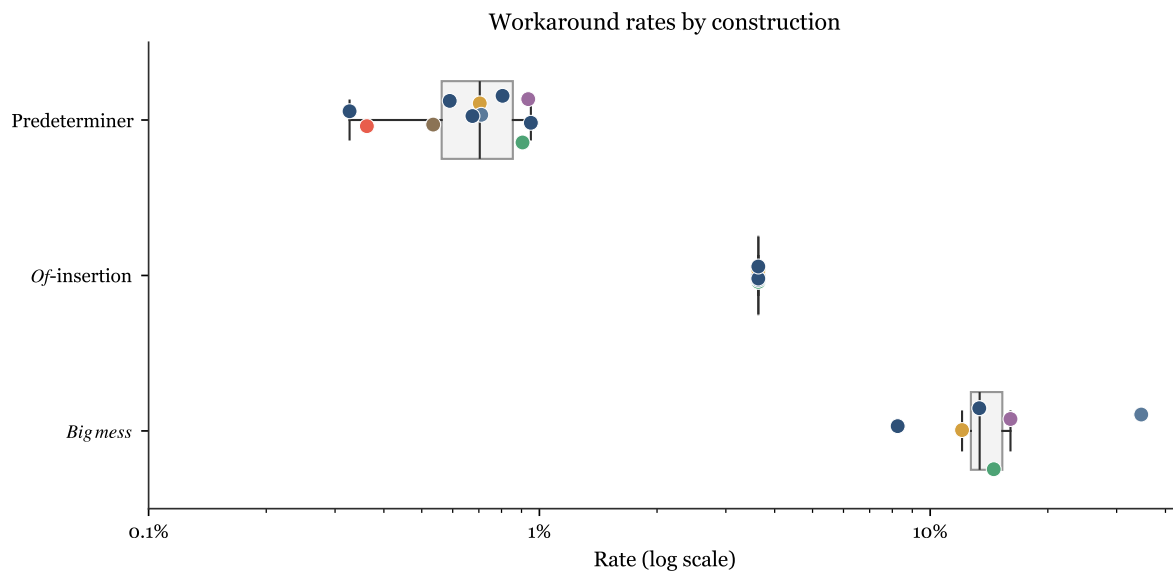


Figure 3: Posterior means and 95% intervals for workaround rates (log scale).

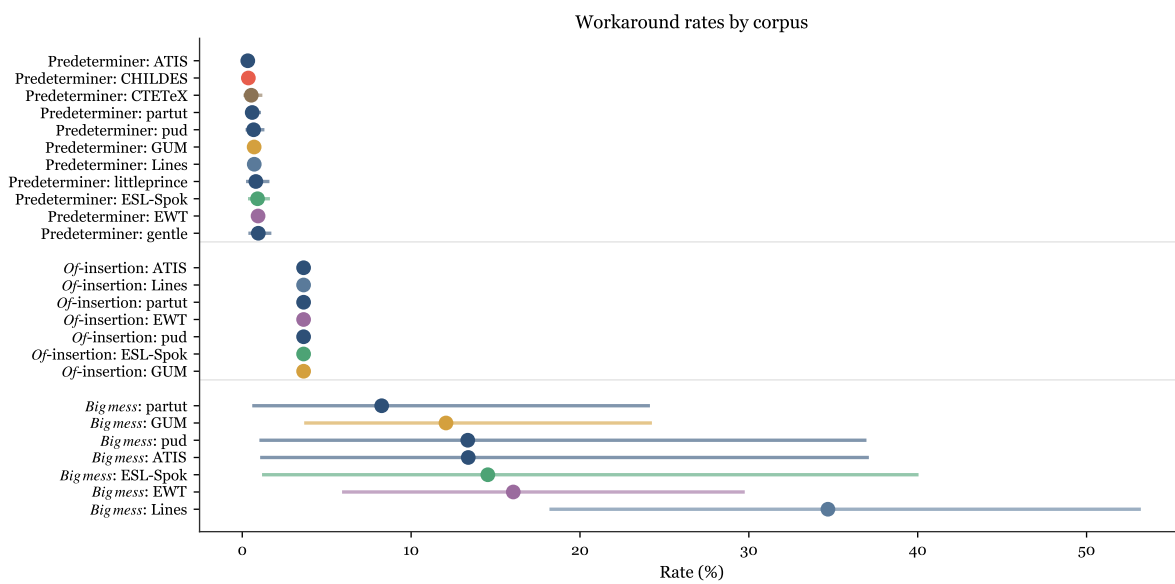


Figure 4: Workaround rates by corpus. Points show posterior means; lines show 95% credible intervals.

Corpus	<i>big mess</i> (%)	<i>of</i> -insertion (%)	predet- <i>a</i> (%)
atis	13.384 [1.159, 37.008]	3.632 [3.519, 3.745]	0.327 [0.121, 0.639]
chldes	–	–	0.362 [0.224, 0.533]
ctetex	–	–	0.535 [0.170, 1.091]
esl	–	–	–
eslpok	14.543 [1.271, 39.950]	3.633 [3.519, 3.748]	0.905 [0.440, 1.538]
ewt	16.050 [6.010, 29.659]	3.633 [3.520, 3.747]	0.936 [0.712, 1.194]
gentle	–	–	0.951 [0.461, 1.624]
gum	12.065 [3.767, 24.161]	3.633 [3.520, 3.747]	0.704 [0.502, 0.940]
gumreddit	–	–	–
lines	34.683 [18.291, 53.115]	3.632 [3.519, 3.746]	0.710 [0.465, 1.007]
littleprince	–	–	0.805 [0.324, 1.510]
partut	8.256 [0.691, 24.043]	3.632 [3.520, 3.746]	0.590 [0.283, 1.008]
pronouns	–	–	–
pud	13.355 [1.114, 36.866]	3.633 [3.519, 3.747]	0.674 [0.294, 1.213]

Table 2: Posterior mean and 95% credible interval for workaround rates. Rates are percent of opportunities.

Configuration	k (tokens)	N (opportunities)	Rate
All det–head arcs exceeding $k=4$	1,229	86,098	1.4%
...with verb or crossing	279	86,098	0.3%
fronted det arcs exceeding $k=4$	12	1,228	1.0%
...with verb or crossing	2	1,228	0.2%

Table 3: Raw counts for determiner–head discontinuity. The final row – fronted determiners with verb or crossing – is the LBE-like configuration; it is near-zero.

the verb flag – false positives rather than genuine LBE-like discontinuities.² The true count of LBE-like configurations is effectively zero.

Figure 5 makes the funnelling visually explicit: from 86,098 determiner–head arcs, we narrow to 1,228 fronted det arcs, then to 12 with distance exceeding $k=4$, and finally to 2 with verb or crossing – both of which are false positives.

A natural objection is that NPs are simply islands, full stop. But English does tolerate extraction from NP-internal positions – specifically, from PP complements of nouns. The corpora contain examples like *Who’s that a picture of?* and *What is that a picture of?*, where the fronted form has been extracted from the *of*-complement of *picture*, stranding the preposition inside the NP. These aren’t marginal or archaic; they are ordinary conversational questions. The contrast is instructive: PP-complement extraction exists at non-trivial rates, while determiner–head discontinuity stays at zero. If NPs are “islands”, English happily swims to the PP shoreline; it just will not pry the determiner off the head in the interior. What’s special isn’t NP in general but the determiner–head packaging unit

²In other words, the verb/crossing flag can’t tell a *moving van* from a van that moves. The heuristic is doing syntax, not semantics; the vans are doing what vans do.

Corpus	fused-head (%)	fronted det verb/cross (%)
atis	57.209 [54.784, 59.635]	0.131 [0.030, 0.306]
childes	88.475 [87.328, 89.589]	0.202 [0.060, 0.427]
ctetex	70.213 [47.569, 88.379]	0.179 [0.041, 0.416]
esl	–	–
eslspok	78.353 [67.750, 87.482]	0.178 [0.040, 0.412]
ewt	89.470 [87.543, 91.264]	0.218 [0.064, 0.462]
gentle	95.890 [90.533, 99.022]	0.178 [0.043, 0.414]
gum	93.897 [92.404, 95.239]	0.173 [0.041, 0.400]
gumreddit	–	–
lines	92.200 [89.632, 94.421]	0.176 [0.041, 0.404]
littleprince	89.662 [80.043, 96.432]	0.178 [0.040, 0.411]
partut	96.529 [93.814, 98.461]	0.179 [0.041, 0.418]
pronouns	–	–
pud	95.643 [90.785, 98.756]	0.179 [0.040, 0.414]

Table 4: Posterior mean and 95% credible interval for fused-head rates and fronted determiner verb/cross rates. Rates are percent of opportunities.

specifically. Figure 6 visualises this contrast.

Table 4, Table 5, and Figure 7 show the posterior rates for fused-head choices and det–head tails. Figure 8 shows the full posterior distributions, making visible how the fronted det verb/cross posteriors pile up at zero while headless rates vary substantially by register. Figure 9 presents the headless rates in forest-plot format, with corpus-specific estimates and a pooled diamond. Figure 10 complements those rates with det–head distance distributions for headed arcs only, and Figure 11 shows that workaround constructions also preserve tight det–head packaging.

The fused-head and pied-piping channels are in direct competition: when a speaker fronts an fronted determiner, they either omit the head (fused-head) or bring it along (pied-piping). Figure 12 plots these rates against each other by corpus. The near-perfect negative correlation confirms that these are genuine alternatives: corpora that favour headless construals correspondingly disfavour headed ones, and vice versa.

These numbers don’t imply impossibility. They imply that LBE-like configurations, as currently operationalised, are exceedingly rare relative to their opportunity sets, while alternative packaging is available and frequent. In-situ fronted determiners (echo questions, embedded interrogatives) are sparse in these written-dominant corpora, but the handful that appear show the same tight det–head packaging – consistent with the main finding, though too few to estimate rates reliably.

7 MEASUREMENT ERROR AND SENSITIVITY

Any corpus search can misfire: it may flag tokens that don’t actually instantiate the target construction (false positives), or miss tokens that do (false negatives). To quantify this, a stratified random sample of 286 rows was drawn from the $k=4$ output – covering flagged hits, near-misses, and clear non-hits – and each row was manually annotated for whether it genuinely instantiated the target pattern. This yields two error rates: the FALSE DISCOVERY RATE (FDR), the proportion of flagged hits that are

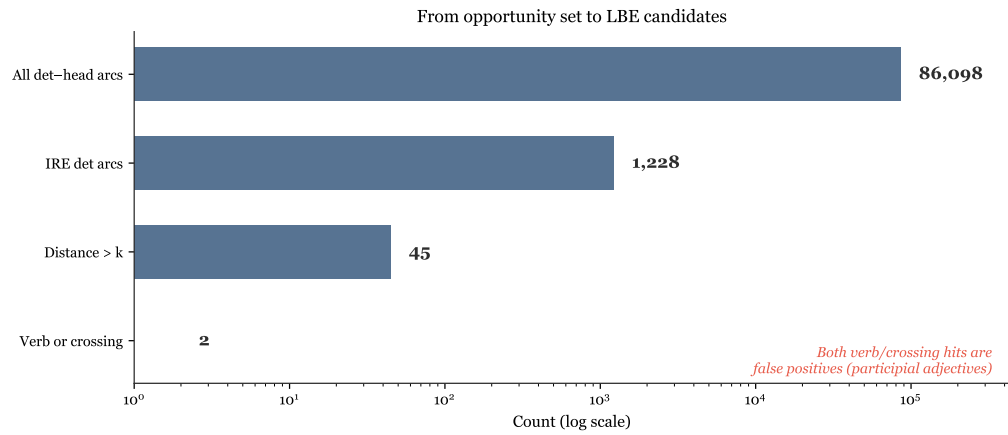


Figure 5: From opportunity set to LBE candidates. The funnel narrows from 86,098 det-head arcs to 2 verb/crossing cases – both false positives (participial adjectives).

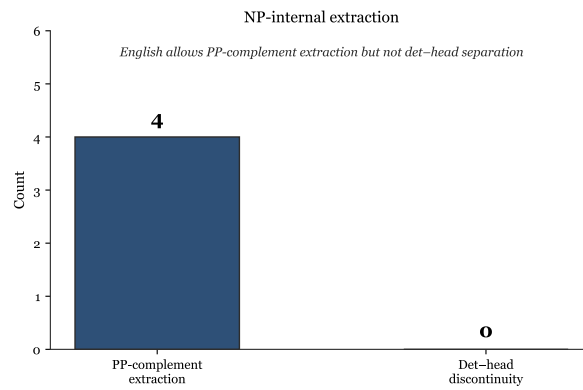


Figure 6: NP-internal extraction: English allows PP-complement extraction but not det-head separation.

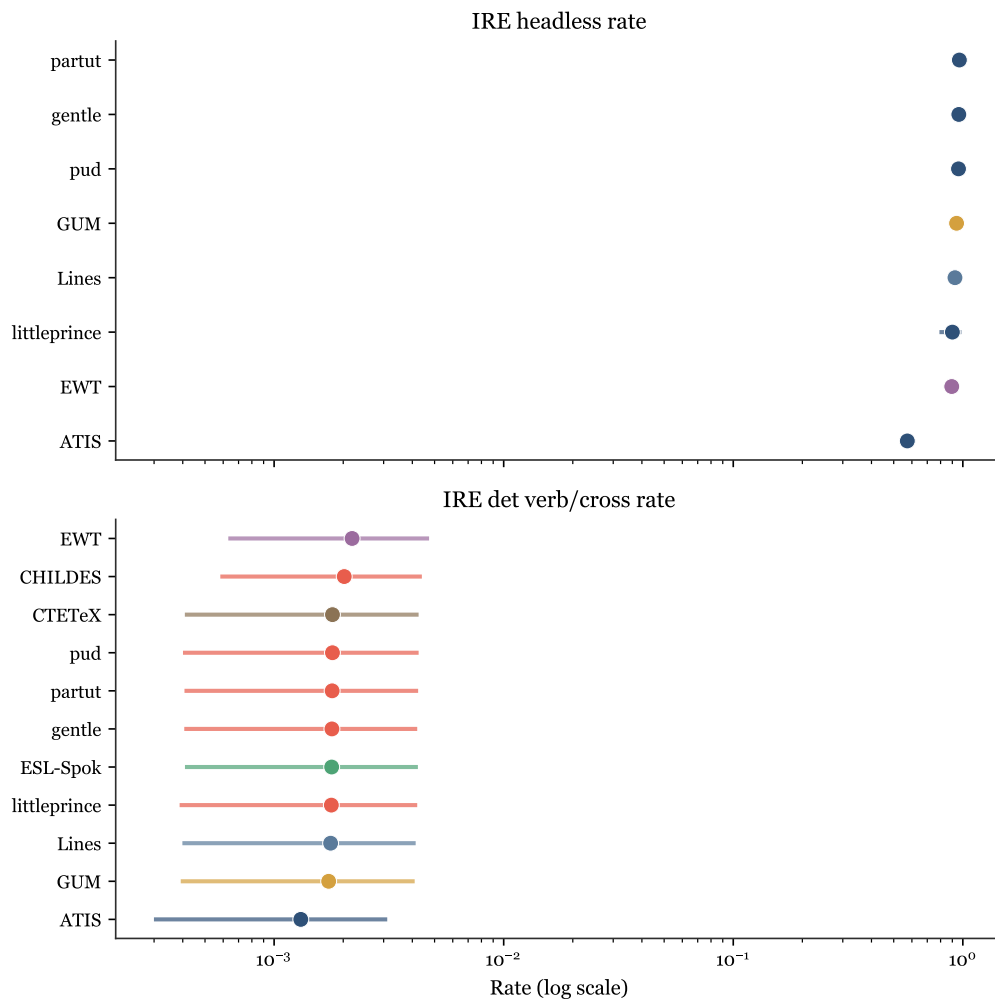


Figure 7: Posterior means and 95% intervals for fused-head rates and fronted determiner verb/cross rates.

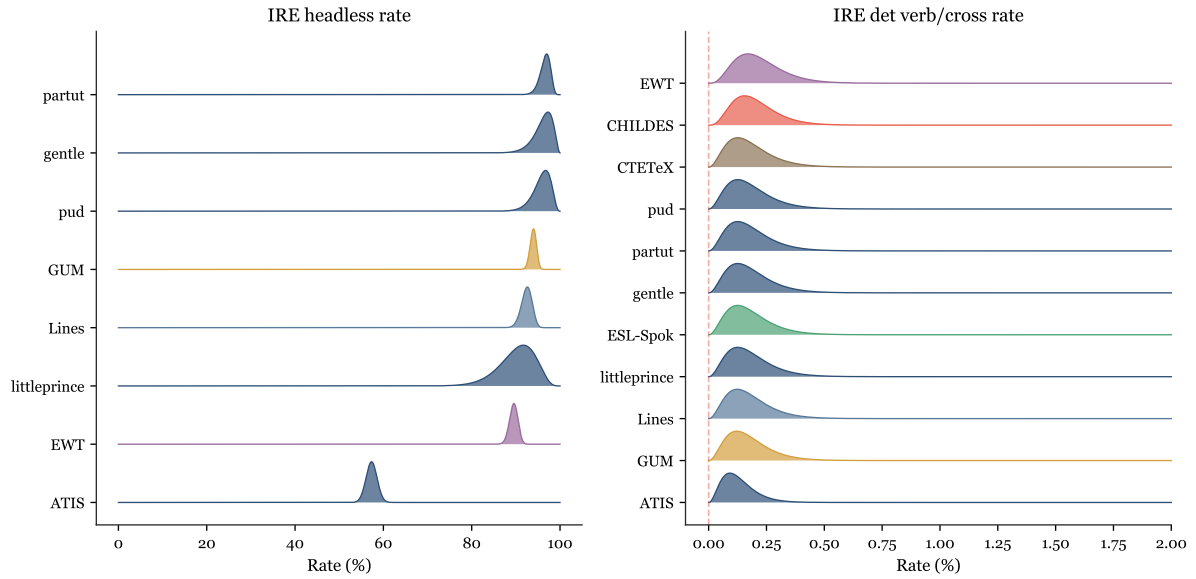


Figure 8: Full posterior distributions for fused-head rates (left) and fronted det verb/cross rates (right). The verb/cross posteriors are concentrated at zero; headless rates show substantial register variation.

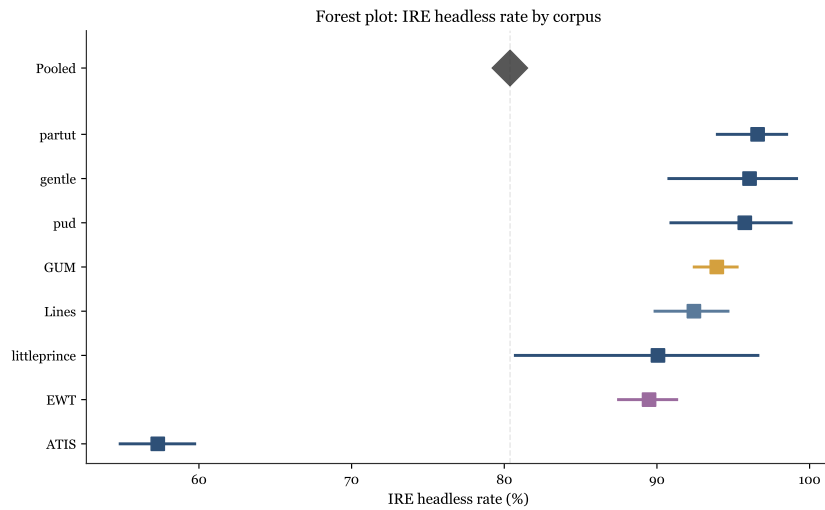


Figure 9: Forest plot: fused-head rate by corpus. Squares show corpus-specific posterior means with 95% intervals; the diamond shows the pooled estimate.

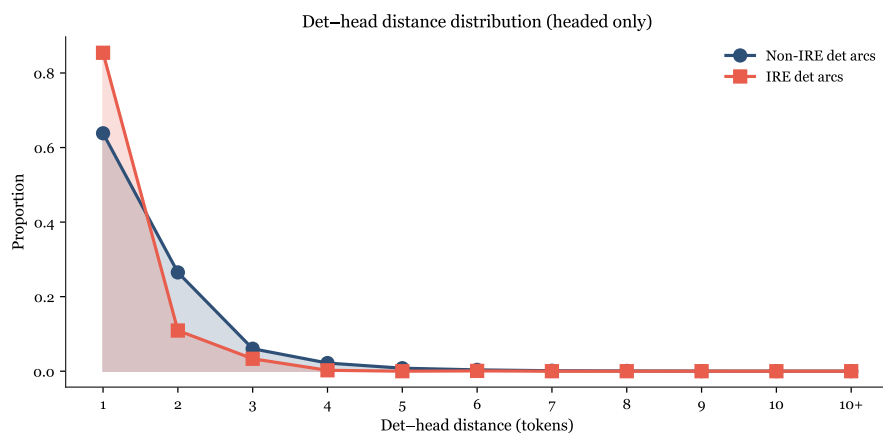


Figure 10: Det-head distance distribution for non-fronted determiner arcs versus fronted determiner arcs (headed only).

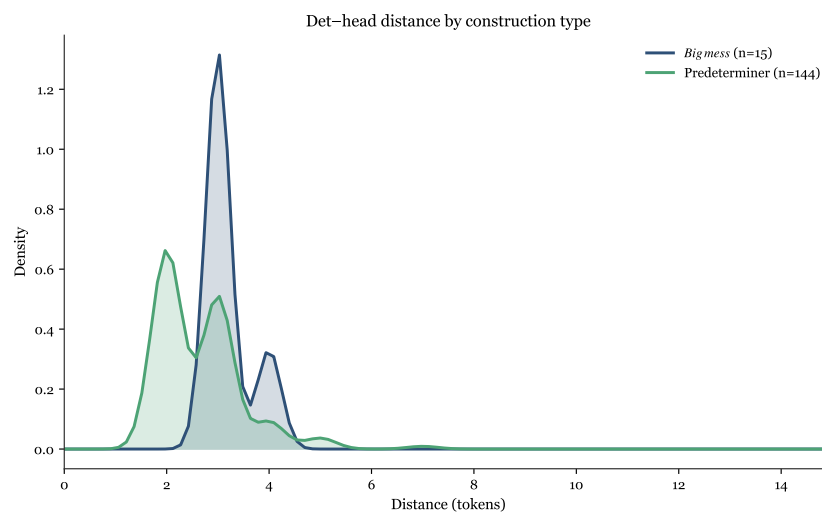


Figure 11: Det-head distance by construction type. Workaround constructions also show tight packaging.

Corpus	det-arc tail (%)	det-arc verb/cross (%)
atis	0.781 [0.557, 1.047]	0.037 [0.004, 0.105]
childes	0.088 [0.046, 0.144]	0.021 [0.004, 0.050]
ctetex	5.819 [4.455, 7.360]	1.813 [1.136, 2.642]
esl	0.694 [0.520, 0.893]	0.153 [0.079, 0.250]
eslspok	0.314 [0.090, 0.678]	0.197 [0.040, 0.479]
ewt	2.294 [2.093, 2.507]	0.569 [0.468, 0.679]
gentle	1.858 [1.191, 2.677]	0.208 [0.041, 0.506]
gum	2.095 [1.893, 2.303]	0.473 [0.379, 0.575]
gumreddit	1.661 [1.024, 2.444]	0.624 [0.285, 1.100]
lines	0.795 [0.632, 0.978]	0.118 [0.061, 0.193]
littleprince	0.506 [0.108, 1.178]	0.379 [0.075, 0.919]
partut	1.673 [1.330, 2.055]	0.326 [0.186, 0.504]
pronouns	0.417 [0.014, 1.461]	0.214 [0.003, 0.826]
pud	1.463 [0.997, 2.008]	0.257 [0.094, 0.502]

Table 5: Posterior mean and 95% credible interval for det–head tail rates. Rates are percent of opportunities.

wrong, and the FALSE OMISSION RATE (FOR), the proportion of non-hits that should have been flagged. The observed FDR is 0.0% (no false positives); the observed FOR is 2.4%. With a $\text{Beta}(1,1)$ prior these correspond to posterior means of 0.6% and 3.2%. The false negatives are concentrated in exclamatives (*what a N*) plus a single *big mess* token, which makes a uniform correction conservative for aggregate rates.

The error model is used as a sensitivity analysis rather than a primary estimate. The posterior on the induced true-hit counts is reported per corpus, which lets readers map back from rates to concrete counts. These adjustments affect the workaround rates, not the fronted determiner det–head tail bound; the LBE proxy is already dominated by the scarcity of candidate configurations.

8 ROBUSTNESS CHECKS

The det–head tail is stable across threshold choices. Re-running with $k = 2$ and $k = 3$ leaves the fronted determiner verb/cross upper bounds below 0.5% in every corpus (max 0.42%). A corpus hold-out posterior predictive check flags a small set of corpus-specific extremes (notably ATIS fused-head rates, LINES *big mess*, and CTETEX/CHILDES det-arc tails), which points to register-specific variation rather than a global misfit. Threshold-specific validation favours $k = 4$: it preserves zero false discoveries while lowering the false omission rate from 14.5% ($k = 2$) to 2.4% ($k = 4$).

Figure 13 summarises the $k = 2/3/4$ robustness checks, and Table 6 lists the lowest posterior predictive tail areas from the corpus hold-out check (diagnostic only, not a threshold test). Figure 14 shows that the posteriors are insensitive to prior choice: uniform $\text{Beta}(1,1)$, Jeffreys $\text{Beta}(0.5,0.5)$, and a skeptical $\text{Beta}(1,100)$ yield nearly identical posteriors for the near-zero fronted det verb/cross rate. For workaround rates with more data, the posteriors converge regardless of prior.

A final check addresses the obvious question: can the data distinguish “LBE truly impossible” from “LBE rare but possible”? With zero genuine cases out of 1,228 opportunities, the 95% upper bound on the true rate is 0.24% – about 1 in 400. If LBE occurred even that rarely, a sample this size

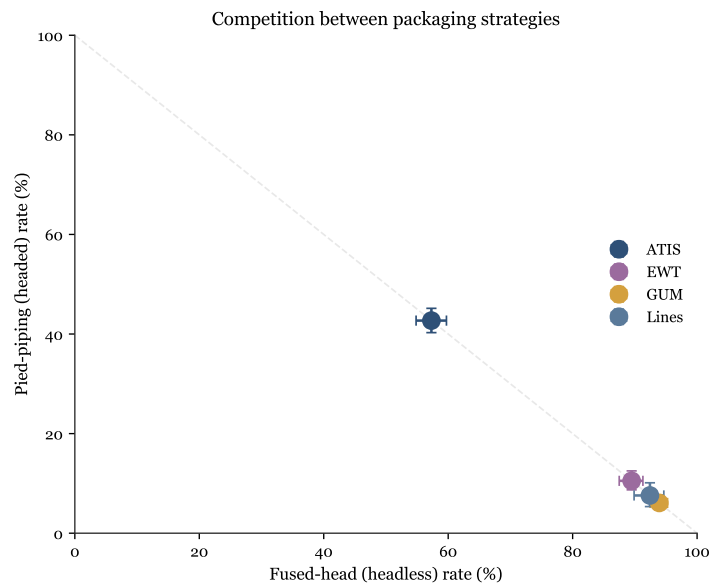


Figure 12: Competition between packaging strategies. Fused-head (headless) rate versus pied-piping (headed) rate by corpus. The diagonal dashed line marks perfect tradeoff.

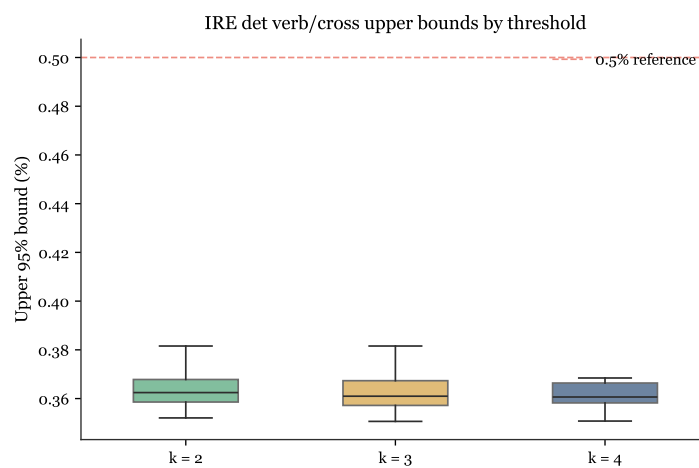


Figure 13: Upper bounds on fronted determiner verb/cross rates by distance threshold.

Rate	Corpus	k/n	Tail area
interrog./rel. headless (int/rel)	atis	898 / 1572	0.000
<i>big mess</i>	lines	8 / 18	0.000
fused-head	atis	898 / 1574	0.000
det-arc verb/cross	ctetex	20 / 920	0.001
det-arc tail	ctetex	57 / 920	0.002
det-arc tail	childes	11 / 13739	0.024
predet- <i>a</i>	atis	1 / 1089	0.039
predet- <i>a</i>	childes	16 / 5042	0.061

Table 6: Lowest posterior predictive tail areas from corpus hold-out checks (diagnostic only).

would have a 95% chance of catching at least one case. It caught none. The sample size is sufficient to detect LBE if it occurred at modestly low rates; zero hits isn’t compatible with “rare but possible” unless “rare” means below 0.1%. Figure 15 visualises this: panel A shows where the true rate probably lies given zero hits; panel B shows how unlikely zero hits would be if LBE were actually possible; panel C shows the detection power – how likely we’d be to catch LBE if it existed at various rates.

9 LIMITATIONS AND FUTURE WORK

The Bayesian story rests on two empirical claims: that learners encounter the relevant evidence, and that the workaround family captures the relevant competitor set. Neither is fully demonstrated here.

On the evidence side, the corpora are predominantly written. The determiner–head packaging claim – that even parentheticals can’t interrupt – is strongest if it holds in speech, not just in edited punctuation conventions. Spoken corpora with time-aligned transcripts or prosodic annotation could measure pause and disfluency incidence between determiner and head versus other NP-internal positions. If determiner–head turns out to be especially boundary-resistant even in disfluent speech, the prosodic-tightness argument is strengthened; if not, the claim needs to be reframed as a production preference rather than a structural impossibility.

On the competitor set, the current inventory may be incomplete. Partitives (*which of these books*, *how many of the students*) occupy a similar communicative niche to pied-piped fronted NPs but route the dependency through a PP. Kind/sort/type-of constructions (*what kind of book*) are another contiguity-preserving alternative. Clefts and pseudo-clefts (*Whose book was it that you read?*) externalise the packaging problem entirely. If these competitors are frequent, the denominator story changes: the absence of LBE is supported not just by pied-piping and fused-head, but by a broader family of schemas that all preserve determiner–head contiguity.

The fused-head proxy (UD PRON vs DET tags) is convenient but imperfect. A gold-annotated sample of fronted strings, coded for fused-head versus headed construal and for local ambiguity, would validate the parsing-bias claim directly.

Finally, the denominators here are global: all determiner–head arcs, all fronting tokens. A more precise Bayesian account would condition on the constructional niche – the contexts where an LBE analysis would be a plausible competitor, not just any determiner. Refining the opportunity set to fronting environments specifically would make the learner’s competition set explicit.

A promising next step would connect this distributional story to the discourse-function tradition

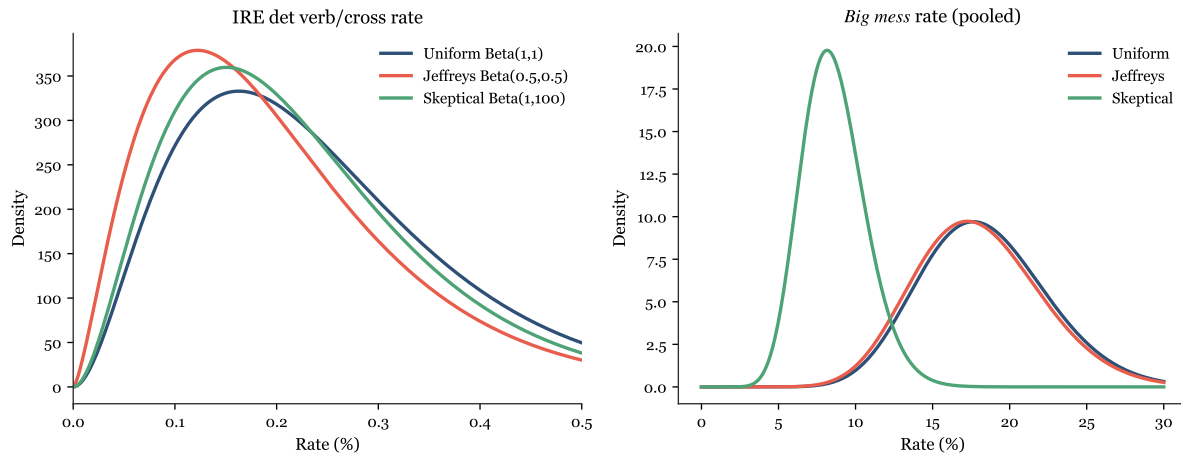


Figure 14: Results don't depend on starting assumptions. Left: fronted det verb/cross rate (near-zero regardless of assumptions). Right: *big mess* rate (stable across different priors). When the data is this clear, the starting point doesn't matter.

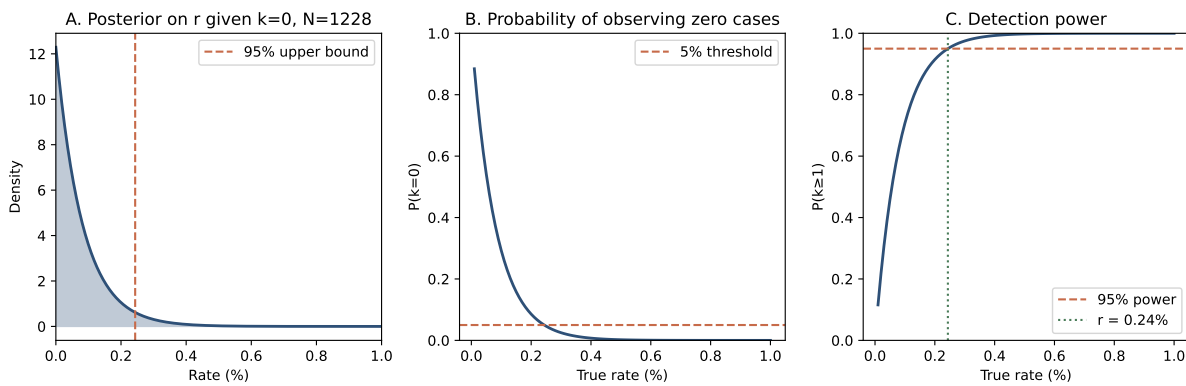


Figure 15: Distinguishing “impossible” from “rare”. (A) Where the true rate probably lies, given zero hits. (B) How likely zero hits would be at various true rates. (C) Detection power: how likely we'd catch at least one case.

(Cuneo & Goldberg, 2023). Within the fronting/degree opportunity set, what determines whether speakers pied-pipe a headed NP, go headless, or choose a workaround? If head omission rises with discourse recoverability and pied-piping rises with head novelty – while det–head discontinuity remains absent regardless – that would link the equilibrium story to discourse pressure. The prediction is sharp: even when the discourse conditions that Cuneo and Goldberg identify as favourable to extraction are met, English should still route around det–head separation.

10 CONCLUSION

Why can't English speakers say **Whose did you read book?* Not because grammar forbids it. Because nobody does it, nobody ever has, and nothing in the input teaches them to.

The explanation has three parts – distributional, psycholinguistic, and functional.

DISTRIBUTIONAL: LBE-like configurations are absent from the data. Across 1,228 fronting opportunities, not a single genuine case of determiner–head discontinuity appears. Meanwhile, workarounds – pied-piping, *big mess*, *of*-insertion, predeterminer modifiers – occur at non-trivial rates. A learner encounters abundant evidence for contiguous packaging and zero evidence for discontinuity.

PSYCHOLINGUISTIC: fused-head parsing commits listeners early. When English speakers hear a fronted *whose*, *which*, or *what*, their default parse treats it as a complete NP. This commitment creates a garden-path effect if a bare noun appears later: the listener expected the dependency to be closed, but the speaker is asking for reanalysis. The sharp unacceptability of LBE reflects the processing cost of that repair.

FUNCTIONAL: the equilibrium is stable because contiguity-preserving constructions have established success conditions. Pied-piping, fused-head construals, and the workaround family all meet communicative needs without separating determiner from head. LBE would offer no new communicative capability – its semantic work is already done by the alternatives – so there is no pressure favouring its emergence. The gap persists not because grammar prohibits LBE, but because LBE has no niche.

The account measures what English speakers actually do, and finds that the alternatives are entrenched while LBE-like configurations are essentially absent. A learner acquiring English has zero evidence for discontinuity and abundant evidence for packaging.

What the analysis documents is the synchronic state: a self-perpetuating equilibrium. No tokens, so no learning. No learning, so no production. No production, so no tokens. This explains the stability of the gap but not its origin. Ultimately, a functional explanation is still needed – some pressure from parsing, production, or discourse that made LBE costly in a way that its competitors were not. Jackendoff's IP-Break Hypothesis identifies a likely structural path for this story: a prosodic bottleneck through which such pressure manifests. If that functional pressure operated in an earlier stage of English, it would have produced the statistical situation now observed. Once established, the equilibrium maintains itself, even without ongoing functional pressure.

But the equilibrium is *contingent*, not necessary. A constraint-based account would say LBE *cannot* emerge in English – there is a grammatical principle blocking it. The functionalist account says only that it *has not* – the social and statistical conditions have never favored it. This is a distinguishing prediction.

Imagine Homer Simpson saying, “Whose did I eat sandwich?” And it's just perfectly placed, as the Simpsons writers so often manage. It gets a big laugh. It's memorable. Fans imitate it, not as

mockery but as in-group signalling. Could it spread? *D'oh! Worst. X. Ever.* did. Why not LBE? The meaning of Homer's question is obvious; the form–meaning pairing is intact. What's missing is only the right environment. Homer doesn't just provide new evidence; he spawns a new discourse community. Fans who adopt the usage aren't just updating a rate parameter; they are signalling membership in a community whose data should be conditioned separately. Within that mixture component, the effective base rate of LBE is no longer zero. It is an Everettian split:³ the innovation instantiates a new component, and the splitting event requires a learner who can identify the community as distinct. The grammar is not resisting; there is no grammar there to resist. There is only convention, and convention can change. Because Bayes!

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³In Hugh Everett's many-worlds interpretation of quantum mechanics, the universe branches at each measurement event. The metaphor here: a new discourse community branches off from the mainstream, carrying its own distributional facts.

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